



Cybersecurity Academy: Configuring URL Filtering on a PA-220 Firewall For a Small Office/Home Office

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Purpose

The purpose of this lab is to configure a PA-220 firewall to filter uniform resource locators (URLs) or links for an elementary school and allow administrators to bypass the filter with an override page. The firewall will first have to be configured for a small office/home office (SOHO) so that it can provide internet connectivity and obtain internet connectivity to Palo Alto's servers. This configuration will allow the firewall to receive updates on URL categories and other security updates on the PA-220.

Background Information

Network administrators may want to filter network traffic from a local company or office. One such way to do this is through URL filtering. URL filtering filters network traffic through the link that a website has (for example, www.example.com). The PA-220 will check all network traffic leaving the local network. In each piece of data leaving the network the URL of the data is checked against a list of URL categories that Palo Alto defines. These URL categories are constantly updated by Palo Alto, so that whenever new threats appear, the network traffic will be filtered from these threats. Network administrators may also want to filter network traffic that is not just malicious but also doesn't promote productivity. For example, in elementary schools, video game websites and related-network traffic would be blocked because it doesn't promote an environment conducive to learning.

One may notice that this filtering could be bypassed by typing the direct address of a location on the internet rather than the URL of the location. But the PA-220 can also be configured to filter IP addresses, keeping the devices on the network from accessing malicious websites.

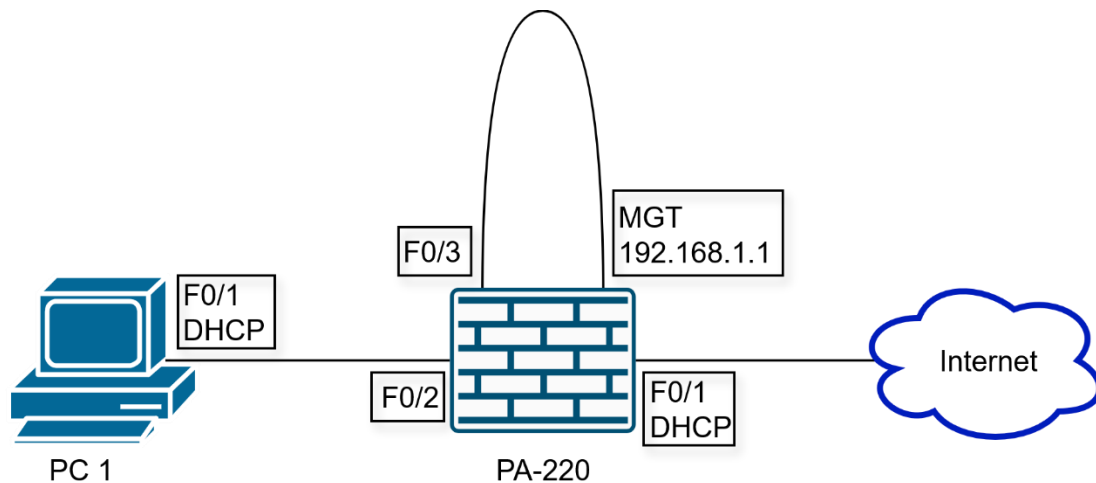
One other problem that may arise is that a network admin needs to access a site that is blocked for either testing purposes or as a one-time check. Palo Alto has a setting available which can create a popup page for blocked websites, which can then allow admins to enter a password in to *override* the block, allowing access. This may be important when access to a website is needed immediately, rather than having to wait for the firewall to be reconfigured to allow the website.

The PA-220, made by The Palo Alto Networks®, is a small NGFW made for small businesses and organizations. It has 8 ethernet ports, two power adapters for redundancy, a console port, and a management port. One interesting feature of the PA-220 is that it can detect hostile data even when the data is encrypted by TLS/SSL by decrypting that data. In addition to that, the PA-220 can integrate with Palo Alto Networks' cloud services. This allows the PA-220 to stop known attacks and malicious URLs from affecting devices on the network. There are many other notable features that the PA-220 has, but please refer to the PA-220 specs for the rest of them.

Lab Summary

In this lab, administrators will configure the PA-220 firewall to filter network traffic leaving the LAN with URL categories. Additionally, administrators will have to configure an override page that allows bypassing of blocked websites with an admin password.

Network Diagram with IPs



Problems

Throughout the process of configuration, many changes were made, so I decided that it would be good to start over from a small office/home office configuration. But when the factory reset was finished, the firewall was stuck in a boot loop. We then had to replace our firewall and reconfigure SOHO, since the backup of our SOHO configuration was corrupted. For next time, I will remember to load a configuration whenever is needed without having to factory reset.

Layer one issues continued in this lab due to faulty wires, so those wires have been replaced to ensure connectivity for the internet during configuration and testing.

Conclusion

URL filtering configuration is an essential part of filtering traffic and preventing access to malicious websites. Through the steps outlined in this document any small company can configure their firewall to route devices to the internet, ensuring work goes smoothly for employees. Additional security configurations can be pursued in order to further secure the network.

Lab Configuration

PA-220 DASHBOARD ACC MONITOR POLICIES OBJECTS NETWORK **DEVICE**

Setup

- High Availability
- Config Audit
- Password Profiles
- Administrators
- Admin Roles
- Authentication Profile
- Authentication Sequence
- User Identification
- Data Redistribution
- Device Quarantine
- VM Information Sources
- Troubleshooting
- Certificate Management
 - Certificates
 - Certificate Profile
 - OCSP Responder
 - SSL/TLS Service Profile
 - SCEP
 - SSL Decryption Exclusion
 - SSH Service Profile
- Response Pages
- Log Settings
- Server Profiles
 - SNMP Trap
 - Syslog
 - Email
 - HTTP
 - Netflow
 - RADIUS
 - SCP
 - TACACS+
 - LDAP
 - Kerberos
 - SAML Identity Provider
 - Multi Factor Authentication

Management | Operations | Services | Interfaces | Telemetry | **Content-ID** | WildFire | Session | ACE | DLP

URL Filtering

URL Continue Timeout (min) 15
URL Admin Override Timeout (min) 15
URL Admin Lockout Timeout (min) 30
Hold client request for category lookup ☐
Category lookup timeout (sec) 2
PAN-DB Server
Append Ending Token ☒

URL Admin Override

LOCATION	SSL/TLS SERVICE PROFILE	MODE	P
Add			
Delete			

HTTP/2 Settings

Connection Logging ☐

Content Cloud Settings

Service URL: hawkeye.services-edge.paloaltonetwork.com

URL Inline Cloud Categorization

Max Latency (ms) 2500
Allow on Max Latency ☒
Log Traffic Not Scanned ☐

URL Admin Override

Select a secure password

Password

Confirm Password

SSL/TLS Service Profile None

Mode ☒ Transparent ☐ Redirect

OK Cancel

- Addresses
- Address Groups
- Regions
- Dynamic User Groups
- Applications
- Application Groups
- Application Filters
- Services
- Service Groups
- Tags
- Devices
- GlobalProtect
 - HIP Objects
 - HIP Profiles
- External Dynamic Lists
- Custom Objects
 - Data Patterns
 - Spyware
 - Vulnerability
 - URL Category
- Security Profiles
 - Antivirus
 - Anti-Spyware
 - Vulnerability Protection
 - URL Filtering**
 - File Blocking
 - WildFire Analysis
 - Data Filtering
 - DoS Protection
- Security Profile Groups
- Log Forwarding
- Authentication
- Decryption
 - Decryption Profile
- SD-WAN Link Management
 - Path Quality Profile
 - SaaS Quality Profile
 - Traffic Distribution Profile
 - Error Correction Profile
- Schedules

NAME	LOCATION
default	Predefined

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

89 items

→

×

☐ CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
Pre-defined Categories		
☐ abortion	alert	alert
☐ abused-drugs	override	block
☐ adult	override	block
☐ AI-code-assistant	override	block
☐ AI-conversational-assistant	override	block
☐ AI-data-and-workflow-optimizer	override	block

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

89 items

→

×

☐ CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐ AI-media-service	override	block
☐ AI-meeting-assistant	override	block
☐ AI-platform-service	override	block
☐ AI-website-generator	override	block
☐ AI-writing-assistant	override	block
☐ alcohol-and-tobacco	override	alert
☐ artificial-intelligence	override	block

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

89 items

→

×

☐ CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐ auctions	alert	alert
☐ business-and-economy	alert	alert
☐ command-and-control	override	block
☐ compromised-website	override	block
☐ computer-and-internet-info	alert	alert
☐ content-delivery-networks	alert	alert
☐ copyright-infringement	override	block

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

89 items

→

×

☐ CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐ cryptocurrency	alert	alert
☐ dating	alert	alert
☐ dynamic-dns	override	block
☐ educational-institutions	alert	alert
☐ encrypted-dns	override	block
☐ entertainment-and-arts	alert	alert
☐ extremism	override	block

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

Q

89 items

→

×

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐	file-converter	alert	alert
☐	financial-services	alert	alert
☐	gambling	override	block
☐	games	override	block
☐	government	alert	alert
☐	grayware	override	block
☐	hacking	override	block
☐	health-and-medicine	alert	alert

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

Q

89 items

→

×

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐	health-and-medicine	alert	alert
☐	high-risk	override	block
☐	home-and-garden	alert	alert
☐	hunting-and-fishing	alert	alert
☐	insufficient-content	override	block
☐	internet-communications-and-telephony	alert	alert
☐	internet-portals	alert	alert
☐	job-search	alert	alert

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

Q

89 items

→

×

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐	job-search	alert	alert
☐	legal	alert	alert
☐	low-risk	alert	alert
☐	malware	override	block
☐	marijuana	override	block
☐	medium-risk	override	block
☐	military	alert	block
☐	motor-vehicles	alert	alert

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

Q

89 items

→

×

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐	motor-vehicles	alert	alert
☐	music	alert	alert
☐	newly-registered-domain	override	block
☐	news	alert	alert
☐	not-resolved	override	block
☐	nudity	override	block
☐	online-storage-and-backup	alert	alert
☐	parked	override	block

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[Check URL Category](#)

OK

Cancel

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

89 items

→

×

☐ CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐ parked	override	block
☐ peer-to-peer	override	block
☐ personal-sites-and-blogs	alert	alert
☐ philosophy-and-political-advocacy	alert	alert
☐ phishing	override	block
☐ private-ip-addresses	override	block
☐ proxy-avoidance-and-anonymizers	override	block
☐ questionable	override	block

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

89 items

→

×

☐ CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐ questionable	override	block
☐ ransomware	override	block
☐ real-estate	alert	alert
☐ real-time-detection	alert	alert
☐ recreation-and-hobbies	alert	alert
☐ reference-and-research	alert	alert
☐ religion	alert	alert
☐ remote-access	alert	alert

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

89 items

☐ CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐ remote-access	alert	alert
☐ scanning-activity	override	block
☐ search-engines	alert	alert
☐ sex-education	alert	alert
☐ shareware-and-freeware	alert	alert
☐ shopping	alert	alert
☐ social-networking	alert	alert
☐ society	+	+

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

89 items

☐ CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐ society	+	+
☐ sports	+	+
☐ stock-advice-and-tools	+	+
☐ streaming-media	+	+
☐ swimsuits-and-intimate-apparel	+	+
☐ training-and-tools	+	+
☐ translation	+	+
☐ travel	+	+

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

89 items

→

×

☐ CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐ training-and-tools	alert	alert
☐ translation	alert	alert
☒ travel	alert	alert
☐ unknown	override	block
☐ weapons	override	block
☐ web-advertisements	alert	alert
☐ web-based-email	alert	alert
☐ web-hosting	alert	alert

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

URL Filtering Profile

Name

elementary school filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

89 items

→

×

☐ CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐ training-and-tools	alert	alert
☐ translation	alert	alert
☒ travel	alert	alert
☐ unknown	override	block
☐ weapons	override	block
☐ web-advertisements	alert	alert
☐ web-based-email	alert	alert
☐ web-hosting	alert	alert

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

PA-220 DASHBOARD ACC MONITOR **POLICIES** OBJECTS NETWORK DEVICE

Security

- NAT
- QoS
- Policy Based Forwarding
- Decryption
- Tunnel Inspection
- Application Override
- Authentication
- DoS Protection
- SD-WAN

NAME	TAGS	TYPE	Source			
			ZONE	ADDRESS	USER	DEVICE
rule1	none	universal	trust	any	any	any
Internet Outgoing	none	universal	Trust-L3	any	any	any
intrazone-default	none	intrazone	any	any	any	any
interzone-default	none	interzone	any	any	any	any

Security Policy Rule

General | Source | Destination | Application | Service/URL Category | **Actions** | Usage

Action Setting

Action: **Allow**

☐ Send ICMP Unreachable

Profile Setting

Profile Type: Profiles

Antivirus: None

Vulnerability Protection: None

Anti-Spyware: None

URL Filtering: **Elementary School Filtering**

File Blocking: None

Data Filtering: None

WildFire Analysis: None

Log Setting

☐ Log at Session Start

☒ Log at Session End

Log Forwarding: None

Other Settings

Schedule: None

QoS Marking: None

☐ Disable Server Response Inspection

OK Cancel

PA-220 DASHBOARD ACC MONITOR POLICIES OBJECTS NETWORK **DEVICE**

Setup

Management | Operations | Services | Interfaces | Telemetry | **Content-ID** | WildFire | Session | ACE | DLP

URL Filtering

URL Continue Timeout (min): 15

URL Admin Override Timeout (min): 15

URL Admin Lockout Timeout (min): 30

Hold client request for category lookup: ☐

Category lookup timeout (sec): 2

PAN-DB Server

Append Ending Token: ☒

URL Admin Override

LOCATION	SSL/TLS SERVICE PROFILE	MODE	PROPERTIES
----------	-------------------------	------	------------

URL Filtering

URL Continue Timeout (min)

URL Admin Override Timeout (min)

URL Admin Lockout Timeout (min)

☒ Hold client request for category lookup

Category lookup timeout (sec)

PAN-DB Server

Cloud list separated by commas

☒ Append Ending Token

PA-220 DASHBOARD ACC MONITOR POLICIES OBJECTS NETWORK **DEVICE**

Device Certificates | Default Trusted Certificate Authorities

NAME	SUBJECT	ISSUER	CA	KEY
------	---------	--------	----	-----

Generate Certificate

Certificate Type ☒ Local ☐ SCEP

Certificate Name

Common Name

Signed By

☒ Certificate Authority ☐ Block Private Key Export

OCSP Responder

Cryptographic Settings

Algorithm

Number of Bits

Digest

Expiration (days)

Certificate Attributes

TYPE	VALUE
------	-------

Generate Certificate



Certificate Type ☒ Local ☐ SCEP

Certificate Name

Common Name

IP or FQDN to appear on the certificate

Signed By

☒ Certificate Authority

☐ Block Private Key Export

OCSP Responder

^ Cryptographic Settings

Algorithm

Number of Bits

Digest

Expiration (days)

Certificate Attributes

☐	TYPE	VALUE
--------------------------	------	-------

Generate

Cancel

Device Certificates

Default Trusted Certificate Authorities

	NAME	SUBJECT	ISSUER	CA	KEY	EXPIRES
	RootCA	CN = 192.168.1.254	CN = 192.168.1.254			Oct 27 19:02:17 2025 GMT

Certificate information

Name

RootCA

Subject

/CN=192.168.1.254

Issuer

/CN=192.168.1.254

Not Valid Before

Oct 27 19:02:17 2025 GMT

Not Valid After

Oct 27 19:02:17 2026 GMT

Algorithm

RSA

☒ Certificate Authority

☐ Forward Trust Certificate

☐ Forward Untrust Certificate

☒ Trusted Root CA

Revoke

OK

Cancel

- Setup
 - High Availability
 - Config Audit
 - Password Profiles
 - Administrators
 - Admin Roles
 - Authentication Profile
 - Authentication Sequence
 - User Identification
 - Data Redistribution
 - Device Quarantine
 - VM Information Sources
 - Troubleshooting
- ✓ Certificate Management
 - Certificates
 - Certificate Profile
 - OCSP Responder
 - SSL/TLS Service Profile**
 - SCEP
 - SSL Decryption Exclusion
 - SSH Service Profile
- Response Pages
- Log Settings
- ✓ Server Profiles
 - SNMP Trap
 - Syslog
 - Email
 - HTTP
 - Netflow
 - RADIUS
 - SCP
 - TACACS+
 - LDAP
 - Kerberos
 - SAML Identity Provider
 - Multi Factor Authentication
- ✓ Local User Database
 - Users
 - User Groups
- Scheduled Log Export
- Software
- GlobalProtect Client
- Dynamic Updates
- Plugins
- Licenses
- Support
- Master Key and Diagnostics
- ✓ Policy Recommendation
 - IoT
 - SaaS



NAME

LC



Add

Delete

Clone

PDF/CSV

SSL/TLS Service Profile

Name **Override Service Profile**

Certificate **OverrideCert**

Protocol Settings

Min Version **TLSv1.2**

Max Version **Max**

OK Cancel

PA-220 DASHBOARD ACC MONITOR POLICIES OBJECTS NETWORK **DEVICE**

Setup Management | Operations | Services | Interfaces | Telemetry **Content-ID** WildFire | Session | ACE | DLP

URL Filtering

URL Continue Timeout (min) 5
URL Admin Override Timeout (min) 5
URL Admin Lockout Timeout (min) 5
Hold client request for category lookup ☐
Category lookup timeout (sec) 2
PAN-DB Server
Append Ending Token ☒

URL Admin Override

LOCATION	SSL/TLS SERVICE PROFILE	MODE
☐ PA-220		transparent

+ Add - Delete

URL Admin Override

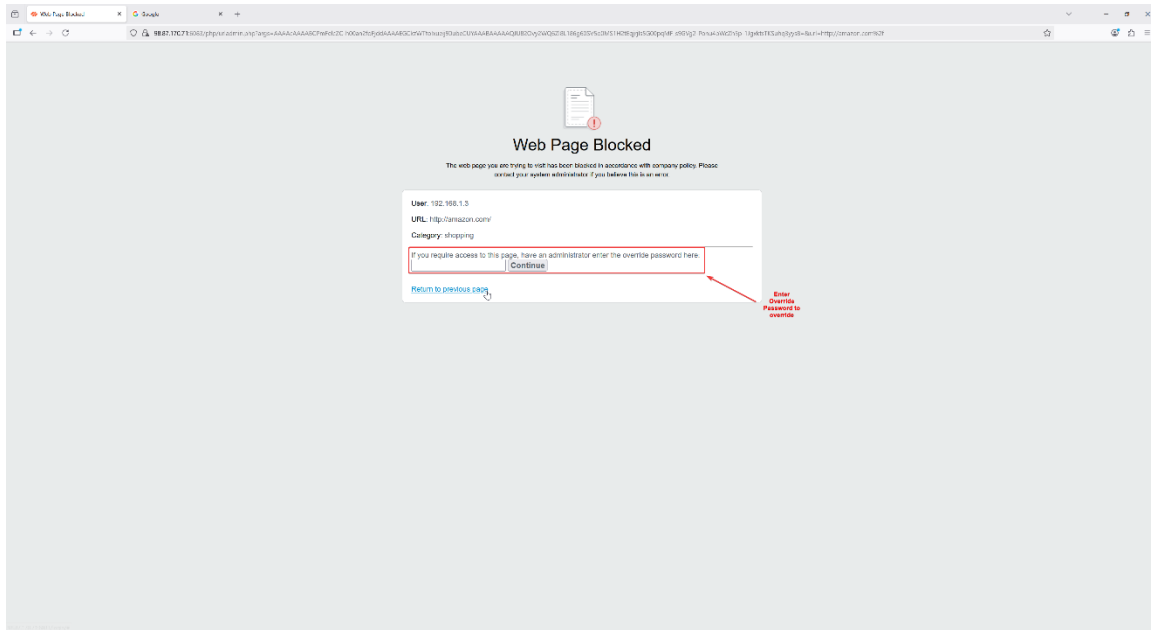
Password **••••••••**

Confirm Password **••••••••**

SSL/TLS Service Profile **Override Service Profile**

Mode ☒ Transparent ☐ Redirect

OK Cancel



Configuration File

```
<?xml version="1.0"?>
<config version="10.2.0" urldb="paloaltonetworks" detail-version="10.2.13">
  <mgt-config>
    <users>
      <entry name="admin">
        <phash>$5$vxxcugxt$RS1Q3Mg7Q0e7hwxfuK1H110dDFRTrJJGxhZFuPTccSB</phash>
        <permissions>
          <role-based>
            <superuser>yes</superuser>
          </role-based>
        </permissions>
      </entry>
    </users>
    <password-complexity>
      <enabled>yes</enabled>
      <minimum-length>8</minimum-length>
    </password-complexity>
  </mgt-config>
  <shared>
    <application/>
    <application-group/>
    <service/>
    <service-group/>
    <botnet>
      <configuration>
        <http>
          <dynamic-dns>
            <enabled>yes</enabled>
            <threshold>5</threshold>
          </dynamic-dns>
          <malware-sites>
            <enabled>yes</enabled>
            <threshold>5</threshold>
          </malware-sites>
          <recent-domains>
            <enabled>yes</enabled>
```

```

    <threshold>5</threshold>
  </recent-domains>
  <ip-domains>
    <enabled>yes</enabled>
    <threshold>10</threshold>
  </ip-domains>
  <executables-from-unknown-sites>
    <enabled>yes</enabled>
    <threshold>5</threshold>
  </executables-from-unknown-sites>
</http>
<other-applications>
  <irc>yes</irc>
</other-applications>
<unknown-applications>
  <unknown-tcp>
    <destinations-per-hour>10</destinations-per-hour>
    <sessions-per-hour>10</sessions-per-hour>
    <session-length>
      <maximum-bytes>100</maximum-bytes>
      <minimum-bytes>50</minimum-bytes>
    </session-length>
  </unknown-tcp>
  <unknown-udp>
    <destinations-per-hour>10</destinations-per-hour>
    <sessions-per-hour>10</sessions-per-hour>
    <session-length>
      <maximum-bytes>100</maximum-bytes>
      <minimum-bytes>50</minimum-bytes>
    </session-length>
  </unknown-udp>
</unknown-applications>
</configuration>
<report>
  <topn>100</topn>
  <scheduled>yes</scheduled>
</report>
</botnet>
<certificate>
  <entry name="RootCA">
    <subject-hash>8a2896e2</subject-hash>
    <issuer-hash>8a2896e2</issuer-hash>
    <not-valid-before>Oct 27 19:02:17 2025 GMT</not-valid-before>
    <issuer>/CN=192.168.1.254</issuer>
    <not-valid-after>Oct 27 19:02:17 2026 GMT</not-valid-after>
    <common-name>192.168.1.254</common-name>
    <expiry-epoch>1793127737</expiry-epoch>
    <ca>yes</ca>
    <subject>/CN=192.168.1.254</subject>
    <public-key>-----BEGIN CERTIFICATE-----
MIIC2zCCAcOgAwIBAgIU50kKwmZdy8eirQ2/yXsK2YSG/u4wDQYJKoZIhvcNAQEL
BQAwGDEwMBQGA1UEAwwNMTkyLjE2OC4xLjI1ND AeFw0yNTEwMjc0TAyMTdaFw0y
NjEwMjc0TAyMTdaMBGxFjAUBgNVBAMMDTE5Mi4xNjguMS4yNTQwggEiMA0GCSqG
SIb3DQEBAQUAA4IBDwAwggEKAoIBAQCkjh5shseyoIaIrk764X3FjZvzdrGDyP
KxiteP+Jv1J7fBPNa2QGjWZhCUEDaW5tJ0SK53T/jxDV/yrrsz+LU08qkjiy7GJA
onET8qvbtdqD8z3CiDWQ2+F/Amk2ZrwrHhHwjSNFxpHCgCrMGmMns1lAk/KSODf
ZyNEJEZg5F1DTtK+N59HUVHLwErOyjVYNq9m3hAZD0A2ELFTnzUN1y5+zwH2ZnqY
WnFBujmv7pxRguRaycU0fIk5IOCGhM3mpV2H3AAVpxYjzWkJeaDvjK49Gyg4gdJS
ZJSxML6De17hiaMS8aKkUmHZwo5nUIAGMqFLHL33zzUlyNA81kbAgMBAAGjHTAb
MAwGA1UdEwQFMAMBAf8wCwYDVVR0PBAQDAGIEMA0GCSqGSIb3DQEBCwUAA4IBAQCW
zzAdr8E8x70MhnZKSo0cdz545Wf89Lj1KUGcVmHV5ORACQoT0BMW+hYD790Srcw4

```

```
bQ96BF4gnLfLAer0hpF8nQC1FqKqKstLPWWDhey5MnvLitQsf6F0eio1B5BvTRIh
v3FjbuaIsnDQ09E+Y1FTXsFSyYkEwn75rwZzL803+WEO0Ceghd+pmtBnbP1fw00Q
XdZWfVqrFzp80jRQir6w+gWtNp6tCFf12hJMTxSUwpAfK7g4+AaRWbGw72SQUc7
+9YR847N7nzx7m7n0ngE9lgnLg2qIsrUSfxT0jzWiTzTk4rFPVso+K5Hbypvi
lkU8nahNTnEopkAgrCWP
-----END CERTIFICATE-----
</public-key>
  <algorithm>RSA</algorithm>
  <private-key>-
AQ==olrIZ+bjM0JRfLLTk2kJmXH6Kw=wfl5Zc+wUAak+B7CnAVY6NdutYwtt67XvF+yGHFGS+vmQugtSrA952X
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</entry>
<entry name="OverrideCert">
  <subject-hash>8a2896e2</subject-hash>
  <issuer-hash>8a2896e2</issuer-hash>
  <not-valid-before>Oct 27 19:10:13 2025 GMT</not-valid-before>
  <issuer>/CN=192.168.1.254</issuer>
  <not-valid-after>Oct 27 19:10:13 2026 GMT</not-valid-after>
  <common-name>192.168.1.254</common-name>
  <expiry-epoch>1793128213</expiry-epoch>
  <ca>yes</ca>
  <subject>/CN=192.168.1.254</subject>
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-----END CERTIFICATE-----

</public-key>

<algorithm>RSA</algorithm>

<private-key>-

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</entry>

</certificate>

<ssl-decrypt>

<trusted-root-CA>

<member>RootCA</member>

</trusted-root-CA>

</ssl-decrypt>

<ssl-tls-service-profile>

<entry name="Override Service Profile">

<protocol-settings>

<min-version>tls1-2</min-version>

<max-version>max</max-version>

</protocol-settings>

<certificate>OverrideCert</certificate>

</entry>

</ssl-tls-service-profile>

</shared>

<devices>

<entry name="localhost.localdomain">

<network>

<interface>

<ethernet>

<entry name="ethernet1/1">

<layer3>

<ndp-proxy>

<enabled>no</enabled>

</ndp-proxy>

<sdwan-link-settings>

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        <upstream-nat>
            <enable>no</enable>
            <static-ip/>
        </upstream-nat>
        <enable>no</enable>
    </sdwan-link-settings>
    <lldp>
        <enable>no</enable>
    </lldp>
    <dhcp-client/>
</layer3>
<link-state>up</link-state>
</entry>
<entry name="ethernet1/2">
    <layer2>
        <lldp>
            <enable>no</enable>
        </lldp>
    </layer2>
</entry>
<entry name="ethernet1/3">
    <layer2>
        <lldp>
            <enable>no</enable>
        </lldp>
    </layer2>
</entry>
</ethernet>
<loopback>
    <units/>
</loopback>
<vlan>
    <units>
        <entry name="vlan.10">
            <ipv6>
                <neighbor-discovery>
                    <router-advertisement>
                        <enable>no</enable>
                    </router-advertisement>
                </neighbor-discovery>
            </ipv6>
            <ndp-proxy>
                <enabled>no</enabled>
            </ndp-proxy>
            <adjust-tcp-mss>
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            </adjust-tcp-mss>
            <ip>
                <entry name="192.168.1.254"/>
            </ip>
        </entry>
    </units>
</vlan>
<tunnel>
    <units/>
</tunnel>
</interface>
<vlan>
    <entry name="Vlan Object">
        <virtual-interface>
            <interface>vlan.10</interface>
        </virtual-interface>
    </entry>
</vlan>

```



```

    </virtual-interface>
    <interface>
        <member>ethernet1/2</member>
        <member>ethernet1/3</member>
    </interface>
</entry>
</vlan>
<virtual-wire/>
<profiles>
    <monitor-profile>
        <entry name="default">
            <interval>3</interval>
            <threshold>5</threshold>
            <action>wait-recover</action>
        </entry>
    </monitor-profile>
</profiles>
<ike>
    <crypto-profiles>
        <ike-crypto-profiles>
            <entry name="default">
                <encryption>
                    <member>aes-128-cbc</member>
                    <member>3des</member>
                </encryption>
                <hash>
                    <member>sha1</member>
                </hash>
                <dh-group>
                    <member>group2</member>
                </dh-group>
                <lifetime>
                    <hours>8</hours>
                </lifetime>
            </entry>
        </ike-crypto-profiles>
        <ipsec-crypto-profiles>
            <entry name="default">
                <esp>
                    <encryption>
                        <member>aes-128-cbc</member>
                        <member>3des</member>
                    </encryption>
                    <authentication>
                        <member>sha1</member>
                    </authentication>
                </esp>
                <dh-group>group2</dh-group>
                <lifetime>
                    <hours>1</hours>
                </lifetime>
            </entry>
        </ipsec-crypto-profiles>
        <global-protect-app-crypto-profiles>
            <entry name="default">
                <encryption>
                    <member>aes-128-cbc</member>
                </encryption>
                <authentication>
                    <member>sha1</member>
                </authentication>
            </entry>
        </global-protect-app-crypto-profiles>
    </crypto-profiles>
</ike>

```

```

        </entry>
    </global-protect-app-crypto-profiles>
</crypto-profiles>
</ike>
<qos>
    <profile>
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            <class-bandwidth-type>
                <mbps>
                    <class>
                        <entry name="class1">
                            <priority>real-time</priority>
                        </entry>
                        <entry name="class2">
                            <priority>high</priority>
                        </entry>
                        <entry name="class3">
                            <priority>high</priority>
                        </entry>
                        <entry name="class4">
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                        </entry>
                        <entry name="class5">
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                        </entry>
                        <entry name="class6">
                            <priority>low</priority>
                        </entry>
                        <entry name="class7">
                            <priority>low</priority>
                        </entry>
                        <entry name="class8">
                            <priority>low</priority>
                        </entry>
                    </class>
                </mbps>
            </class-bandwidth-type>
        </entry>
    </profile>
</qos>
<virtual-router>
    <entry name="default">
        <protocol>
            <bgp>
                <enable>no</enable>
                <dampening-profile>
                    <entry name="default">
                        <cutoff>1.25</cutoff>
                        <reuse>0.5</reuse>
                        <max-hold-time>900</max-hold-time>
                        <decay-half-life-reachable>300</decay-half-life-reachable>
                        <decay-half-life-unreachable>900</decay-half-life-unreachable>
                        <enable>yes</enable>
                    </entry>
                </dampening-profile>
                <routing-options>
                    <graceful-restart>
                        <enable>yes</enable>
                    </graceful-restart>
                </routing-options>
            </bgp>
        </protocol>
    </entry>
</virtual-router>

```

```

    <rip>
      <enable>no</enable>
    </rip>
    <ospf>
      <enable>no</enable>
    </ospf>
    <ospfv3>
      <enable>no</enable>
    </ospfv3>
  </protocol>
  <interface>
    <member>ethernet1/1</member>
    <member>vlan.10</member>
  </interface>
  <ecmp>
    <algorithm>
      <ip-modulo/>
    </algorithm>
  </ecmp>
</entry>
</virtual-router>
<dhcp>
  <interface>
    <entry name="vlan.10">
      <server>
        <option>
          <dns>
            <primary>inherited</primary>
            <secondary>inherited</secondary>
          </dns>
          <wins>
            <primary>inherited</primary>
            <secondary>inherited</secondary>
          </wins>
          <nis>
            <primary>inherited</primary>
            <secondary>inherited</secondary>
          </nis>
          <ntp>
            <primary>inherited</primary>
            <secondary>inherited</secondary>
          </ntp>
          <inheritance>
            <source>ethernet1/1</source>
          </inheritance>
          <lease>
            <unlimited/>
          </lease>
          <dns-suffix>inherited</dns-suffix>
          <pop3-server>inherited</pop3-server>
          <smtp-server>inherited</smtp-server>
          <gateway>192.168.1.254</gateway>
          <subnet-mask>255.255.255.0</subnet-mask>
        </option>
        <ip-pool>
          <member>192.168.1.2-192.168.1.253</member>
        </ip-pool>
        <mode>enabled</mode>
      </server>
    </entry>
  </interface>

```

```

    </dhcp>
</network>
<deviceconfig>
  <system>
    <ip-address>192.168.1.1</ip-address>
    <netmask>255.255.255.0</netmask>
    <update-server>updates.paloaltonetworks.com</update-server>
    <update-schedule>
      <threats>
        <recurring>
          <every-30-mins>
            <at>0</at>
            <action>download-and-install</action>
          </every-30-mins>
        </recurring>
      </threats>
    </update-schedule>
    <timezone>US/Pacific</timezone>
    <service>
      <disable-telnet>yes</disable-telnet>
      <disable-http>yes</disable-http>
    </service>
    <hostname>PA-220</hostname>
    <device-telemetry>
      <device-health-performance>yes</device-health-performance>
      <product-usage>yes</product-usage>
      <threat-prevention>yes</threat-prevention>
      <region>Americas</region>
    </device-telemetry>
    <type>
      <static/>
    </type>
    <default-gateway>192.168.1.254</default-gateway>
    <dns-setting>
      <servers>
        <primary>9.9.9.9</primary>
        <secondary>1.1.1.1</secondary>
      </servers>
    </dns-setting>
  </system>
  <setting>
    <config>
      <rematch>yes</rematch>
    </config>
    <management>
      <hostname-type-in-syslog>FQDN</hostname-type-in-syslog>
    </management>
    <ctd>
      <url-coach-timeout>5</url-coach-timeout>
      <url-admin-timeout>5</url-admin-timeout>
      <url-lockout-timeout>5</url-lockout-timeout>
      <hold-client-request>yes</hold-client-request>
      <url-wait-timeout>15</url-wait-timeout>
    </ctd>
  </setting>
</deviceconfig>
<vsys>
  <entry name="vsys1">
    <application/>
    <application-group/>
    <zone>

```

```

<entry name="trust">
  <network>
    <virtual-wire/>
  </network>
</entry>
<entry name="untrust">
  <network>
    <virtual-wire/>
  </network>
</entry>
<entry name="Trust-L3">
  <network>
    <layer3>
      <member>vlan.10</member>
    </layer3>
  </network>
</entry>
<entry name="Trust-L2">
  <network>
    <layer2>
      <member>ethernet1/2</member>
      <member>ethernet1/3</member>
    </layer2>
  </network>
</entry>
<entry name="Untrust-L3">
  <network>
    <layer3>
      <member>ethernet1/1</member>
    </layer3>
  </network>
</entry>
</zone>
<service/>
<service-group/>
<schedule/>
<rulebase>
  <security>
    <rules>
      <entry name="rule1" uuid="bdd9df4d-4bf7-4daa-a64f-943800043b7b">
        <from>
          <member>trust</member>
        </from>
        <to>
          <member>untrust</member>
        </to>
        <source>
          <member>any</member>
        </source>
        <destination>
          <member>any</member>
        </destination>
        <service>
          <member>any</member>
        </service>
        <application>
          <member>any</member>
        </application>
        <action>allow</action>
        <log-end>yes</log-end>
      </entry>
    </rules>
  </security>
</rulebase>

```

```
d3026f19d1c6">    <entry name="Internet Outgoing" uuid="1608d8ec-8082-4b0c-b000-
```

```
    <to>
      <member>Untrust-L3</member>
    </to>
    <from>
      <member>Trust-L3</member>
    </from>
    <source>
      <member>any</member>
    </source>
    <destination>
      <member>any</member>
    </destination>
    <source-user>
      <member>any</member>
    </source-user>
    <category>
      <member>any</member>
    </category>
    <application>
      <member>any</member>
    </application>
    <service>
      <member>application-default</member>
    </service>
    <source-hip>
      <member>any</member>
    </source-hip>
    <destination-hip>
      <member>any</member>
    </destination-hip>
    <action>allow</action>
    <description>All traffic to the internet.</description>
    <profile-setting>
      <profiles>
        <url-filtering>
          <member>Elementary School Filtering</member>
        </url-filtering>
      </profiles>
    </profile-setting>
    <log-start>no</log-start>
    <log-end>yes</log-end>
```

```
  </entry>
```

```
</rules>
```

```
</security>
```

```
<nat>
```

```
  <rules>
```

```
    <entry name="Internet Outgoing" uuid="0037b321-6d5e-424c-ba66-7fe03d6e09a2">
```

```
      <source-translation>
        <dynamic-ip-and-port>
          <interface-address>
            <interface>ethernet1/1</interface>
          </interface-address>
        </dynamic-ip-and-port>
      </source-translation>
      <to>
        <member>Untrust-L3</member>
      </to>
      <from>
```

```

        <member>Trust-L3</member>
    </from>
    <source>
        <member>any</member>
    </source>
    <destination>
        <member>any</member>
    </destination>
    <service>any</service>
    <to-interface>ethernet1/1</to-interface>
</entry>
</rules>
</nat>
</rulebase>
<import>
    <network>
        <interface>
            <member>ethernet1/1</member>
            <member>ethernet1/2</member>
            <member>ethernet1/3</member>
            <member>vlan.10</member>
        </interface>
    </network>
</import>
<address>
    <entry name="192.168.1.254">
        <ip-netmask>192.168.1.254/24</ip-netmask>
    </entry>
    <entry name="IP address">
        <ip-netmask>192.168.1.254/24</ip-netmask>
    </entry>
</address>
<profiles>
    <url-filtering>
        <entry name="Elementary School Filtering">
            <credential-enforcement>
                <mode>
                    <disabled/>
                </mode>
                <log-severity>medium</log-severity>
                <alert>
                    <member>abortion</member>
                    <member>auctions</member>
                    <member>business-and-economy</member>
                    <member>computer-and-internet-info</member>
                    <member>content-delivery-networks</member>
                    <member>cryptocurrency</member>
                    <member>educational-institutions</member>
                    <member>entertainment-and-arts</member>
                    <member>file-converter</member>
                    <member>financial-services</member>
                    <member>government</member>
                    <member>health-and-medicine</member>
                    <member>home-and-garden</member>
                    <member>hunting-and-fishing</member>
                    <member>internet-communications-and-telephony</member>
                    <member>internet-portals</member>
                    <member>job-search</member>
                    <member>legal</member>
                    <member>low-risk</member>
                    <member>military</member>
                </alert>
            </credential-enforcement>
        </entry>
    </url-filtering>
</profiles>

```

```
<member>motor-vehicles</member>
<member>music</member>
<member>news</member>
<member>online-storage-and-backup</member>
<member>parked</member>
<member>personal-sites-and-blogs</member>
<member>philosophy-and-political-advocacy</member>
<member>real-estate</member>
<member>real-time-detection</member>
<member>recreation-and-hobbies</member>
<member>reference-and-research</member>
<member>religion</member>
<member>remote-access</member>
<member>search-engines</member>
<member>sex-education</member>
<member>shareware-and-freeware</member>
<member>social-networking</member>
<member>society</member>
<member>sports</member>
<member>stock-advice-and-tools</member>
<member>streaming-media</member>
<member>swimsuits-and-intimate-apparel</member>
<member>training-and-tools</member>
<member>translation</member>
<member>travel</member>
<member>web-advertisements</member>
<member>web-based-email</member>
<member>web-hosting</member>
</alert>
<block>
  <member>abused-drugs</member>
  <member>adult</member>
  <member>AI-code-assistant</member>
  <member>AI-conversational-assistant</member>
  <member>AI-data-and-workflow-optimizer</member>
  <member>AI-media-service</member>
  <member>AI-meeting-assistant</member>
  <member>AI-platform-service</member>
  <member>AI-website-generator</member>
  <member>AI-writing-assistant</member>
  <member>alcohol-and-tobacco</member>
  <member>artificial-intelligence</member>
  <member>command-and-control</member>
  <member>compromised-website</member>
  <member>copyright-infringement</member>
  <member>dating</member>
  <member>dynamic-dns</member>
  <member>encrypted-dns</member>
  <member>extremism</member>
  <member>gambling</member>
  <member>games</member>
  <member>grayware</member>
  <member>hacking</member>
  <member>high-risk</member>
  <member>insufficient-content</member>
  <member>malware</member>
  <member>marijuana</member>
  <member>medium-risk</member>
  <member>newly-registered-domain</member>
  <member>not-resolved</member>
  <member>nudity</member>
```



```
<member>peer-to-peer</member>
<member>phishing</member>
<member>private-ip-addresses</member>
<member>proxy-avoidance-and-anonymizers</member>
<member>questionable</member>
<member>ransomware</member>
<member>scanning-activity</member>
<member>shopping</member>
<member>unknown</member>
<member>weapons</member>
</block>
</credential-enforcement>
<alert>
  <member>abortion</member>
  <member>auctions</member>
  <member>business-and-economy</member>
  <member>computer-and-internet-info</member>
  <member>content-delivery-networks</member>
  <member>cryptocurrency</member>
  <member>educational-institutions</member>
  <member>entertainment-and-arts</member>
  <member>file-converter</member>
  <member>financial-services</member>
  <member>government</member>
  <member>health-and-medicine</member>
  <member>home-and-garden</member>
  <member>hunting-and-fishing</member>
  <member>internet-communications-and-telephony</member>
  <member>internet-portals</member>
  <member>job-search</member>
  <member>legal</member>
  <member>low-risk</member>
  <member>military</member>
  <member>motor-vehicles</member>
  <member>music</member>
  <member>news</member>
  <member>online-storage-and-backup</member>
  <member>parked</member>
  <member>personal-sites-and-blogs</member>
  <member>philosophy-and-political-advocacy</member>
  <member>real-estate</member>
  <member>real-time-detection</member>
  <member>recreation-and-hobbies</member>
  <member>reference-and-research</member>
  <member>religion</member>
  <member>remote-access</member>
  <member>search-engines</member>
  <member>sex-education</member>
  <member>shareware-and-freeware</member>
  <member>social-networking</member>
  <member>society</member>
  <member>sports</member>
  <member>stock-advice-and-tools</member>
  <member>streaming-media</member>
  <member>swimsuits-and-intimate-apparel</member>
  <member>training-and-tools</member>
  <member>translation</member>
  <member>travel</member>
  <member>web-advertisements</member>
  <member>web-based-email</member>
  <member>web-hosting</member>
```

```

</alert>
<override>
  <member>abused-drugs</member>
  <member>adult</member>
  <member>AI-code-assistant</member>
  <member>AI-conversational-assistant</member>
  <member>AI-data-and-workflow-optimizer</member>
  <member>AI-media-service</member>
  <member>AI-meeting-assistant</member>
  <member>AI-platform-service</member>
  <member>AI-website-generator</member>
  <member>AI-writing-assistant</member>
  <member>alcohol-and-tobacco</member>
  <member>artificial-intelligence</member>
  <member>command-and-control</member>
  <member>compromised-website</member>
  <member>copyright-infringement</member>
  <member>dating</member>
  <member>dynamic-dns</member>
  <member>encrypted-dns</member>
  <member>extremism</member>
  <member>gambling</member>
  <member>games</member>
  <member>grayware</member>
  <member>hacking</member>
  <member>high-risk</member>
  <member>insufficient-content</member>
  <member>malware</member>
  <member>marijuana</member>
  <member>medium-risk</member>
  <member>newly-registered-domain</member>
  <member>not-resolved</member>
  <member>nudity</member>
  <member>peer-to-peer</member>
  <member>phishing</member>
  <member>private-ip-addresses</member>
  <member>proxy-avoidance-and-anonymizers</member>
  <member>questionable</member>
  <member>ransomware</member>
  <member>scanning-activity</member>
  <member>shopping</member>
  <member>unknown</member>
  <member>weapons</member>
</override>
</entry>
</url-filtering>
</profiles>
<profile-group>
  <entry name="Internet">
    <virus>
      <member>default</member>
    </virus>
    <spyware>
      <member>strict</member>
    </spyware>
    <vulnerability>
      <member>strict</member>
    </vulnerability>
    <url-filtering>
      <member>default</member>
    </url-filtering>
  </entry>
</profile-group>

```

```

        </entry>
      </profile-group>
      <url-admin-override>
        <password>-
AQ==jnt+yDgU5gmEHDtcejTndJ4AbnY=HW9KHQ4pnb0snRslqBxSA==</password>
        <mode>
          <transparent/>
        </mode>
        <ssl-tls-service-profile>Override Service Profile</ssl-tls-service-profile>
      </url-admin-override>
    </entry>
  </vsys>
</entry>
</devices>
</config>

```

URL Profile Output

```

admin@PA-220# show profiles url-filtering "Elementary School Filtering"
"Elementary School Filtering" {
  credential-enforcement {
    mode {
      disabled;
    }
    log-severity medium;
    alert [ abortion auctions business-and-economy computer-and-internet-info
content-delivery-networks cryptocurrency educational-institutions
entertainment-and-arts file-converter financial-services government health-
and-medicine home-and-garden hunting-and-fishing internet-communications-and-
telephony internet-portals job-search legal low-risk military motor-vehicles
music news online-storage-and-backup parked personal-sites-and-blogs
philosophy-and-political-advocacy real-estate real-time-detection recreation-
and-hobbies reference-and-research religion remote-access search-engines sex-
education shareware-and-freeware social-networking society sports stock-
advice-and-tools streaming-media swimsuits-and-intimate-apparel training-and-
tools translation travel web-advertisements web-based-email web-hosting];
    block [ abused-drugs adult AI-code-assistant AI-conversational-assistant
AI-data-and-workflow-optimizer AI-media-service AI-meeting-assistant AI-
platform-service AI-website-generator AI-writing-assistant alcohol-and-tobacco
artificial-intelligence command-and-control compromised-website copyright-
infringement dating dynamic-dns encrypted-dns extremism gambling games
grayware hacking high-risk insufficient-content malware marijuana medium-
riskresolved nudity peer-to-peer phishing private-ip-addresses proxy-avoidance-
and-anonymizers questionable ransomware scanning-activity shopping unknown
weapons];
  }
  alert [ abortion auctions business-and-economy computer-and-internet-info
content-delivery-networks cryptocurrency educational-institutions
entertainment-and-arts file-converter financial-services government health-
and-medicine home-and-garden hunting-and-fishing internet-communications-and-
telephony internet-portals job-search legal low-risk military motor-vehicles
music news online-storage-and-backup parked personal-sites-and-blogs
philosophy-and-political-advocacy real-estate real-time-detection recreation-
and-hobbies reference-and-research religion remote-access search-engines sex-
education shareware-and-freeware social-networking society sports stock-
advice-and-tools streaming-media swimsuits-and-intimate-apparel training-and-
tools translation travel web-advertisements web-based-email web-hosting];
}

```

```
    override [ abused-drugs adult AI-code-assistant AI-conversational-assistant  
AI-data-and-workflow-optimizer AI-media-service AI-meeting-assistant AI-  
platform-service AI-website-generator AI-writing-assistant alcohol-and-tobacco  
artificial-intelligence command-and-control compromised-website copyright-  
infringement dating dynamic-dns encrypted-dns extremism gambling games  
grayware hacking high-risk insufficient-content malware marijuana medium-risk  
newly-registered-domain not-resolved nudity peer-to-peer phishing private-ip-  
addresses proxy-avoidance-and-anonymizers questionable ransomware scanning-  
activity shopping unknown weapons];  
}
```